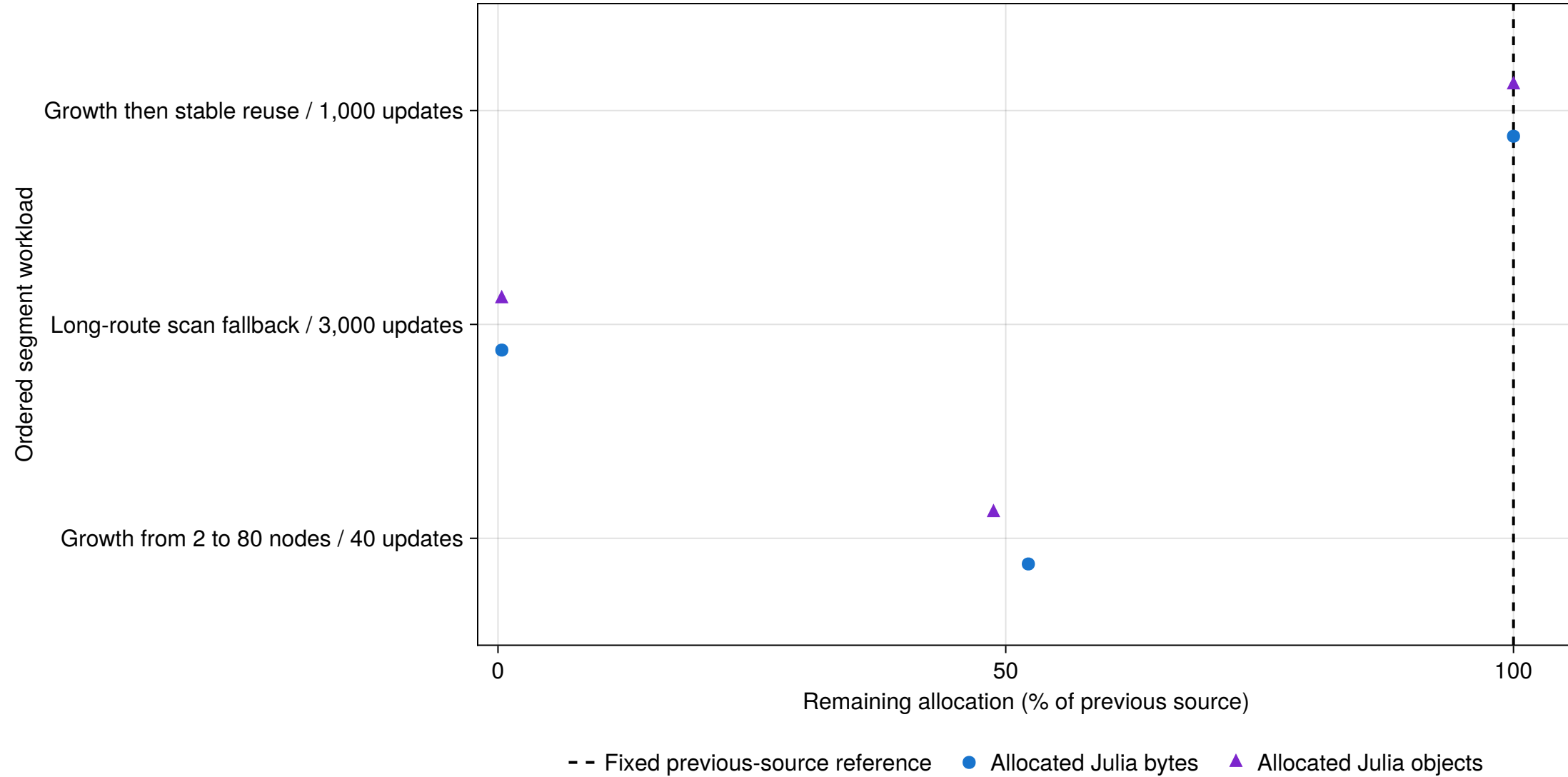


# Actual range-cache sequences: bounded private storage



180,138 exact original segment comparisons; cache creation and growing private storage included. Stable reuse is unchanged.  
No original LR101 cache allocation gain: all nine full paired paths preserve bytes/objects exactly. No controlled speed or quality claim.